BCA201P — LAB SHEET 04

Foundations of Java Development | BCA III Semester | Session 2026–2027

Java program solutions for Questions 1–30. Each program is self-contained; save each program in a separate .java file using its public class name (these examples use non-public classes).

Q1. Student class

```
class Student {
    String name;
    int age;
    static int count = 0;

    Student(String n, int a) {
        String studentName = n;
        int studentAge = a;
        name = studentName;
        age = studentAge;
        count++;
    }

    void display() {
        System.out.println("Name: " + name + ", Age: " + age);
    }

    public static void main(String[] args) {
        Student s1 = new Student("Aman", 20);
        Student s2 = new Student("Riya", 19);
        s1.display();
        s2.display();
        System.out.println("Total students: " + count);
    }
}
```

Q2. Rectangle area

```
class Rectangle {
    double length, breadth;
    static String shapeName = "Rectangle";

    Rectangle(double l, double b) {
        length = l;
        breadth = b;
    }

    void displayArea() {
        double area = length * breadth;
        System.out.println(shapeName + " area: " + area);
    }

    public static void main(String[] args) {
        Rectangle r = new Rectangle(8, 4);
        r.displayArea();
    }
}
```

Q3. Circle area and circumference

```
class Circle {
    double radius;
    static final double PI = 3.14;

    Circle(double r) { radius = r; }

    void display() {
        double area = PI * radius * radius;
        double circumference = 2 * PI * radius;
        System.out.println("Area: " + area);
        System.out.println("Circumference: " + circumference);
    }

    public static void main(String[] args) {
        Circle c = new Circle(5);
        c.display();
    }
}
```

Q4. Employee details

```
class Employee {
    int empId;
    double salary;
    static String companyName = "ABC Ltd.";

    Employee(int id, double pay) {
        empId = id;
        salary = pay;
    }

    void display() {
        int id = empId;
        double pay = salary;
        System.out.println("ID: " + id + ", Salary: " + pay);
        System.out.println("Company: " + companyName);
    }

    public static void main(String[] args) {
        Employee e = new Employee(101, 35000);
        e.display();
    }
}
```

Q5. Book details

```
class Book {
    String title, author;
    double price;
    static String publisher = "ABC Publications";

    Book(String t, String a, double p) {
        title = t; author = a; price = p;
    }

    void display() {
        String bookTitle = title;
        String bookAuthor = author;
        double bookPrice = price;
        System.out.println(bookTitle + " by " + bookAuthor);
        System.out.println("Price: " + bookPrice);
        System.out.println("Publisher: " + publisher);
    }

    public static void main(String[] args) {
        new Book("Java Basics", "R. Sharma", 299).display();
    }
}
```

Q6. Car details

```
class Car {
    String brand;
    double mileage;
    static int wheels = 4;

    Car(String b, double m) { brand = b; mileage = m; }

    void display() {
        String carBrand = brand;
        double carMileage = mileage;
        System.out.println("Brand: " + carBrand);
        System.out.println("Mileage: " + carMileage);
        System.out.println("Wheels: " + wheels);
    }

    public static void main(String[] args) {
        new Car("Toyota", 18.5).display();
    }
}
```

Q7. Bank account deposit

```
class BankAccount {
    long accountNumber;
    double balance;
    static String bankName = "State Bank";

    BankAccount(long number, double amount) {
        accountNumber = number; balance = amount;
    }

    void deposit(double amount) {
        double depositAmount = amount;
        if (depositAmount > 0) balance += depositAmount;
        System.out.println("Balance: " + balance);
    }

    public static void main(String[] args) {
        BankAccount account = new BankAccount(123456, 1000);
        System.out.println("Bank: " + bankName);
        account.deposit(500);
    }
}
```

Q8. Pen details

```
class Pen {
    String color, type;
    static String manufacturer = "Cello";

    Pen(String c, String t) { color = c; type = t; }

    void display() {
        String penColor = color;
        String penType = type;
        System.out.println("Color: " + penColor);
        System.out.println("Type: " + penType);
        System.out.println("Manufacturer: " + manufacturer);
    }

    public static void main(String[] args) {
        new Pen("Blue", "Ballpoint").display();
    }
}
```

Q9. Laptop details

```
class Laptop {
    String brand;
    int RAM;
    static String os = "Windows";

    Laptop(String b, int r) { brand = b; RAM = r; }

    void display() {
        String laptopBrand = brand;
        int laptopRAM = RAM;
        System.out.println("Brand: " + laptopBrand);
        System.out.println("RAM: " + laptopRAM + " GB");
        System.out.println("OS: " + os);
    }

    public static void main(String[] args) {
        new Laptop("Dell", 8).display();
    }
}
```

Q10. Movie details

```
class Movie {
    String name, genre;
    double rating;
    static String industry = "Bollywood";

    Movie(String n, String g, double r) {
        name = n; genre = g; rating = r;
    }

    void display() {
        String movieName = name;
        String movieGenre = genre;
        double movieRating = rating;
        System.out.println(movieName + " (" + movieGenre + ")");
        System.out.println("Rating: " + movieRating);
        System.out.println("Industry: " + industry);
    }

    public static void main(String[] args) {
        new Movie("Example Film", "Drama", 8.0).display();
    }
}
```

Q11. Calculator and operation counter

```
class Calculator {
    static int operationsCount = 0;

    void add(double a, double b) {
        double result = a + b; operationsCount++;
        System.out.println("Addition: " + result);
    }
    void subtract(double a, double b) {
        double result = a - b; operationsCount++;
        System.out.println("Subtraction: " + result);
    }
    void multiply(double a, double b) {
        double result = a * b; operationsCount++;
        System.out.println("Multiplication: " + result);
    }
    void divide(double a, double b) {
        if (b == 0) {
            System.out.println("Cannot divide by zero.");
            return;
        }
        double result = a / b; operationsCount++;
        System.out.println("Division: " + result);
    }

    public static void main(String[] args) {
        Calculator c = new Calculator();
        c.add(10, 5); c.subtract(10, 5);
        c.multiply(10, 5); c.divide(10, 5);
        System.out.println("Operations: " + operationsCount);
    }
}
```

Q12. Student marks average

```
class StudentMarks {
    int marks1, marks2, marks3;
    static String universityName = "ABC University";

    StudentMarks(int a, int b, int c) {
        marks1 = a; marks2 = b; marks3 = c;
    }

    void displayAverage() {
        double total = marks1 + marks2 + marks3;
        double average = total / 3;
        System.out.println("University: " + universityName);
        System.out.println("Average: " + average);
    }

    public static void main(String[] args) {
        new StudentMarks(75, 80, 85).displayAverage();
    }
}
```

Q13. Temperature conversion

```
class Temperature {
    double celsius;
    static String scale = "Celsius";

    Temperature(double c) { celsius = c; }

    void convert() {
        double fahrenheit = (celsius * 9 / 5) + 32;
        double kelvin = celsius + 273.15;
        System.out.println(celsius + " " + scale);
        System.out.println("Fahrenheit: " + fahrenheit);
        System.out.println("Kelvin: " + kelvin);
    }

    public static void main(String[] args) {
        new Temperature(25).convert();
    }
}
```

Q14. Voting eligibility

```
class Person {
    String name;
    int age;
    static String country = "India";

    Person(String n, int a) { name = n; age = a; }

    void checkEligibility() {
        int personAge = age;
        if (personAge >= 18)
            System.out.println(name + " is eligible to vote.");
        else
            System.out.println(name + " is not eligible to vote.");
        System.out.println("Country: " + country);
    }

    public static void main(String[] args) {
        new Person("Aman", 19).checkEligibility();
    }
}
```

Q15. Simple interest

```
class SimpleInterest {
    double principal, rate, time;
    static String bank = "ABC Bank";

    SimpleInterest(double p, double r, double t) {
        principal = p; rate = r; time = t;
    }

    void calculate() {
        double interest = (principal * rate * time) / 100;
        System.out.println("Bank: " + bank);
        System.out.println("Simple interest: " + interest);
    }

    public static void main(String[] args) {
        new SimpleInterest(10000, 5, 2).calculate();
    }
}
```

Q16. Triangle validity and type

```
class Triangle {
    double a, b, c;
    static String type = "Scalene/Isosceles/Equilateral";

    Triangle(double x, double y, double z) {
        a = x; b = y; c = z;
    }

    void check() {
        double x = a, y = b, z = c;
        if (x <= 0 || y <= 0 || z <= 0 ||
            x + y <= z || x + z <= y || y + z <= x) {
            System.out.println("Invalid triangle");
        } else if (x == y && y == z) {
            System.out.println("Equilateral triangle");
        } else if (x == y || y == z || x == z) {
            System.out.println("Isosceles triangle");
        } else {
            System.out.println("Scalene triangle");
        }
        System.out.println("Possible types: " + type);
    }

    public static void main(String[] args) {
        new Triangle(3, 4, 5).check();
    }
}
```

Q17. Student grade

```
class StudentGrade {
    String name;
    int marks;
    static int passingMarks = 40;

    StudentGrade(String n, int m) { name = n; marks = m; }

    void showGrade() {
        int score = marks;
        String grade;
        if (score < passingMarks) grade = "F";
        else if (score >= 90) grade = "A";
        else if (score >= 75) grade = "B";
        else if (score >= 60) grade = "C";
        else grade = "D";
        System.out.println(name + " grade: " + grade);
    }

    public static void main(String[] args) {
        new StudentGrade("Riya", 82).showGrade();
    }
}
```

Q18. Electric bill

```
class ElectricBill {
    int units;
    static double fixedCharge = 50.0;

    ElectricBill(int u) { units = u; }

    void calculate() {
        int consumedUnits = units;
        double energyCharge;
        if (consumedUnits <= 100) energyCharge = consumedUnits * 2.0;
        else if (consumedUnits <= 200)
            energyCharge = 100 * 2.0 + (consumedUnits - 100) * 3.0;
        else
            energyCharge = 100 * 2.0 + 100 * 3.0
                         + (consumedUnits - 200) * 5.0;
        double totalBill = energyCharge + fixedCharge;
        System.out.println("Total bill: Rs. " + totalBill);
    }

    public static void main(String[] args) {
        new ElectricBill(150).calculate();
    }
}
```

Q19. Product discount

```
class Product {
    int id;
    String name;
    double price;
    static double discountRate = 10.0;

    Product(int i, String n, double p) {
        id = i; name = n; price = p;
    }

    void displayFinalPrice() {
        double discount = price * discountRate / 100;
        double finalPrice = price - discount;
        System.out.println(id + ": " + name);
        System.out.println("Final price: Rs. " + finalPrice);
    }

    public static void main(String[] args) {
        new Product(1, "Notebook", 100).displayFinalPrice();
    }
}
```

Q20. Bank withdrawal

```
class BankAccount {
    double balance;
    static String bankCode = "BANK001";

    BankAccount(double b) { balance = b; }

    void withdraw(double amount) {
        double withdrawalAmount = amount;
        if (withdrawalAmount <= 0)
            System.out.println("Enter a valid amount.");
        else if (withdrawalAmount > balance)
            System.out.println("Insufficient balance.");
        else {
            balance -= withdrawalAmount;
            System.out.println("Remaining balance: " + balance);
        }
    }

    public static void main(String[] args) {
        BankAccount account = new BankAccount(5000);
        System.out.println("Bank code: " + bankCode);
        account.withdraw(1200);
    }
}
```

Q21. Palindrome number

```
class Palindrome {
    static int countChecks = 0;

    void check(int number) {
        int original = number;
        int n = number;
        int reverse = 0;
        countChecks++;
        while (n > 0) {
            int digit = n % 10;
            reverse = reverse * 10 + digit;
            n /= 10;
        }
        if (original >= 0 && original == reverse)
            System.out.println(original + " is a palindrome.");
        else
            System.out.println(original + " is not a palindrome.");
    }

    public static void main(String[] args) {
        Palindrome p = new Palindrome();
        p.check(121); p.check(123);
        System.out.println("Checks: " + countChecks);
    }
}
```

Q22. Armstrong number

```
class Armstrong {
    static int totalChecks = 0;

    void check(int number) {
        int n = number;
        int digits = String.valueOf(Math.abs(number)).length();
        int sum = 0;
        totalChecks++;
        while (n > 0) {
            int digit = n % 10;
            sum += (int) Math.pow(digit, digits);
            n /= 10;
        }
        if (number >= 0 && sum == number)
            System.out.println(number + " is an Armstrong number.");
        else
            System.out.println(number + " is not an Armstrong number.");
    }

    public static void main(String[] args) {
        Armstrong a = new Armstrong();
        a.check(153); a.check(123);
        System.out.println("Checks: " + totalChecks);
    }
}
```

Q23. Prime checker

```
class PrimeChecker {
    static int totalPrimeChecks = 0;

    void check(int number) {
        boolean prime = number >= 2;
        totalPrimeChecks++;
        for (int i = 2; i <= number / i; i++) {
            if (number % i == 0) {
                prime = false;
                break;
            }
        }
        System.out.println(number + (prime ? " is prime." : " is not prime."));
    }

    public static void main(String[] args) {
        PrimeChecker p = new PrimeChecker();
        p.check(17); p.check(20);
        System.out.println("Checks: " + totalPrimeChecks);
    }
}
```

Q24. Fibonacci series

```
class Fibonacci {
    static int seriesCount = 0;

    void display(int n) {
        int first = 0, second = 1;
        seriesCount++;
        for (int i = 0; i < n; i++) {
            System.out.print(first + " ");
            int next = first + second;
            first = second;
            second = next;
        }
        System.out.println();
    }

    public static void main(String[] args) {
        Fibonacci f = new Fibonacci();
        f.display(10);
        System.out.println("Series displayed: " + seriesCount);
    }
}
```

Q25. Factorial

```
class Factorial {
    static int totalCalls = 0;

    void calculate(int number) {
        long factorial = 1;
        totalCalls++;
        if (number < 0) {
            System.out.println("Factorial is not defined for negatives.");
            return;
        }
        for (int i = 1; i <= number; i++) factorial *= i;
        System.out.println("Factorial: " + factorial);
    }

    public static void main(String[] args) {
        Factorial f = new Factorial();
        f.calculate(5);
        System.out.println("Calls: " + totalCalls);
    }
}
```

Q26. Add and subtract 2x2 matrices

```
class Matrix {
    static String matrixType = "2x2";

    public static void main(String[] args) {
        int[][] a = {{1, 2}, {3, 4}};
        int[][] b = {{5, 6}, {7, 8}};
        int[][] sum = new int[2][2];
        int[][] difference = new int[2][2];

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                sum[i][j] = a[i][j] + b[i][j];
                difference[i][j] = a[i][j] - b[i][j];
            }
        }

        System.out.println("Matrix type: " + matrixType);
        System.out.println("Addition:");
        for (int[] row : sum) {
            for (int value : row) System.out.print(value + " ");
            System.out.println();
        }
        System.out.println("Subtraction:");
        for (int[] row : difference) {
            for (int value : row) System.out.print(value + " ");
            System.out.println();
        }
    }
}
```

Q27. Reverse a string

```
class StringReverse {
    static int totalReversals = 0;

    void reverse(String text) {
        String reversed = "";
        totalReversals++;
        for (int i = text.length() - 1; i >= 0; i--)
            reversed += text.charAt(i);
        System.out.println("Reversed: " + reversed);
    }

    public static void main(String[] args) {
        StringReverse s = new StringReverse();
        s.reverse("Java");
        System.out.println("Reversals: " + totalReversals);
    }
}
```

Q28. Word counter

```
class WordCounter {
    static String language = "English";

    void countWords(String sentence) {
        String cleaned = sentence.trim();
        int wordCount = cleaned.isEmpty() ? 0 : cleaned.split("\\s+").length;
        System.out.println("Language: " + language);
        System.out.println("Word count: " + wordCount);
    }

    public static void main(String[] args) {
        new WordCounter().countWords("Java is easy to learn");
    }
}
```

Q29. Compare employee salaries

```
class Employee {
    String name;
    double salary;
    static String organization = "ABC Organization";

    Employee(String n, double s) { name = n; salary = s; }

    public static void main(String[] args) {
        Employee e1 = new Employee("Aman", 30000);
        Employee e2 = new Employee("Riya", 35000);
        double salary1 = e1.salary;
        double salary2 = e2.salary;
        System.out.println("Organization: " + organization);
        if (salary1 > salary2)
            System.out.println(e1.name + " earns more.");
        else if (salary2 > salary1)
            System.out.println(e2.name + " earns more.");
        else
            System.out.println("Both salaries are equal.");
    }
}
```

Q30. Library issue and return

```
class Library {
    int booksAvailable;
    static String libraryName = "City Library";

    Library(int books) { booksAvailable = books; }

    void issueBook() {
        int available = booksAvailable;
        if (available > 0) {
            booksAvailable--;
            System.out.println("Book issued.");
        } else {
            System.out.println("No books available.");
        }
        System.out.println("Books left: " + booksAvailable);
    }

    void returnBook() {
        int returnedBooks = 1;
        booksAvailable += returnedBooks;
        System.out.println("Book returned.");
        System.out.println("Books available: " + booksAvailable);
    }

    public static void main(String[] args) {
        Library library = new Library(5);
        System.out.println("Library: " + libraryName);
        library.issueBook();
        library.returnBook();
    }
}
```